PROJECT AQUAFLUSH

Arduino-Based Water-Flow Monitoring for a Miniature Urban Drainage Model

Technical Laboratory Report | BCA144 | May 2025 project

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Figure 1. Miniature roadway and drainage exhibit, retained from the supplied project photographs.

Abstract

AquaFlush is an academic drainage demonstration inspired by flooding concerns in Barangay Tibanga, Iligan City. The recovered Arduino sketch acquires water-flow sensor pulses, estimates flow in liters per hour, and presents the reading on a 16×2 I²C LCD and Serial Monitor. The team integrated these components into a physical scale model. This report describes the available monitoring implementation and records its hardware, source provenance, and development boundaries.

Controller	Input	Outputs
Arduino Uno / ATmega328P	Pulse-output flow sensor; exact model unconfirmed	16×2 I²C LCD; Serial at 9600 baud

Purpose and objectives

Acquire sensor pulses, calculate an estimated flow value, display the reading locally, and integrate the electronics into a miniature drainage model. Additional sensing and automation remain possible future development.

This is a documentation revision based on the supplied archive. Additional physical trials were not available; no new measurement results are included.

1. System and measurement method

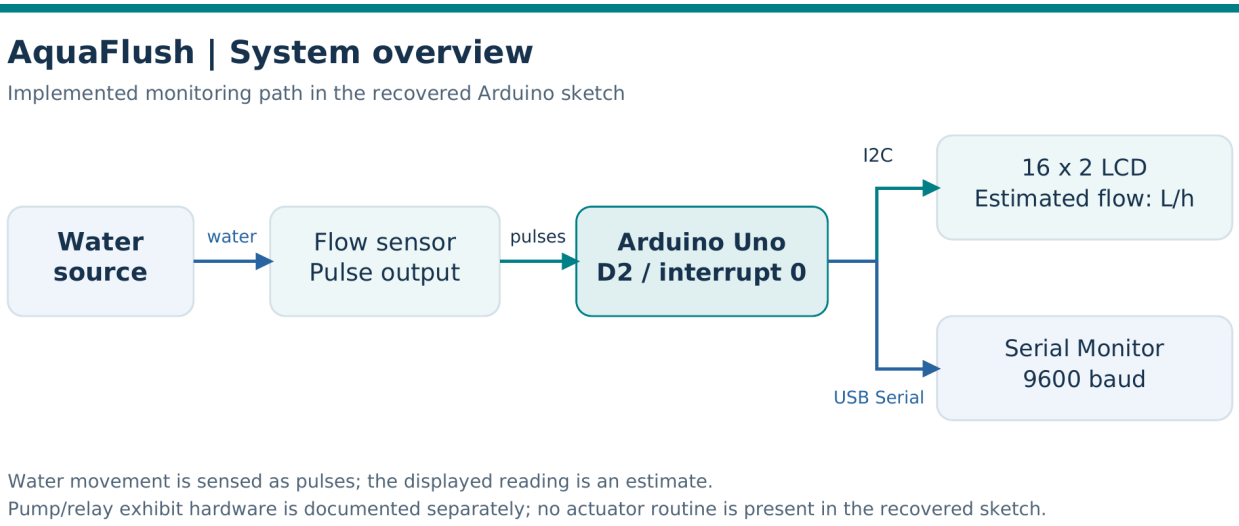


Figure 2. Implemented monitoring path derived from the recovered source.

Pulse-to-flow conversion

A rising edge on D2 increments a shared counter. Roughly once per second, the main loop reads that count, calculates a whole-number flow estimate, resets the count, and updates the LCD and Serial output.

```
l_hour = (flow_frequency * 60 / 7.5);
```

For interval Δt in seconds and a sensor factor k in Hz per L/min, frequency is N / Δt and flow is 60 × N / (Δt × k) L/hour. The sketch assumes Δt ≈ 1 second and k = 7.5. The exact sensor identity and its calibration factor remain unconfirmed.

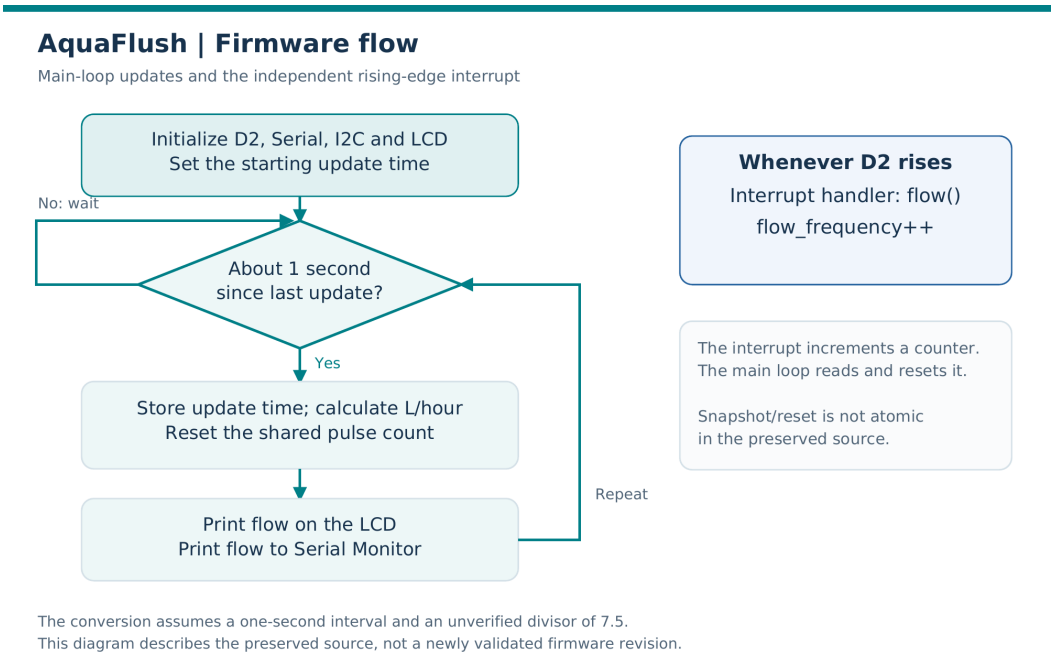


Figure 3. Source-derived main-loop sequence and independent pulse interrupt.

The preserved implementation uses non-atomic shared-counter access, a fixed interval assumption, and a time comparison that is not robust across millis() rollover. A later firmware iteration could address these details; this revision changes documentation only.

2. Hardware and reference wiring

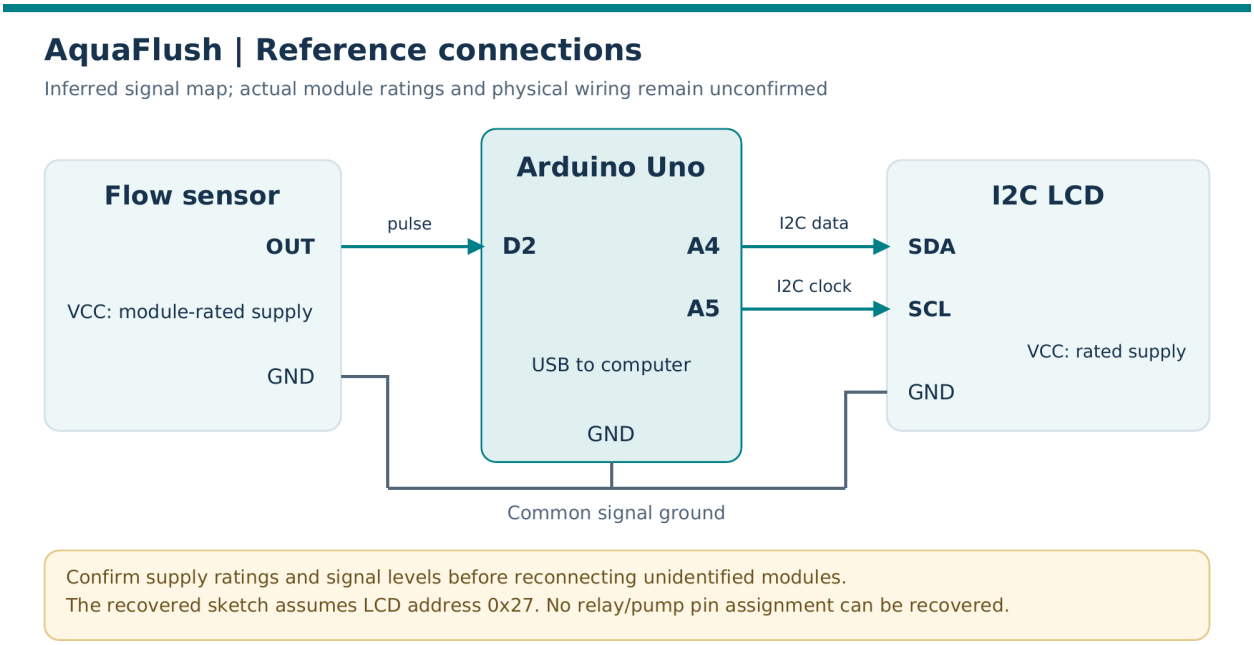


Figure 4. Reference connections inferred from the sketch and Uno pinout; not a verified as-built schematic.

Bill of materials

Component	Qty.	Identification / purpose
Arduino Uno	1	ATmega328P platform; sensor acquisition and outputs
Flow sensor	1	Pulse-output type; exact model unconfirmed
LCD with I²C backpack	1	16×2 characters; code assumes address 0x27
Breadboard / jumpers	1 / set	Prototype connections; exact wire count not recorded
Relay module	1	Identified as 5 V in supplied notes; exhibit hardware
Water pump	1	Shown in exhibit; rating/model not recorded
Container / tubing	Set	Physical water path; dimensions not recorded
Scale drainage model	1	Team-built roadway and drainage exhibit
USB data cable	1 for reuse	Computer upload and Serial access

The original component costs were not retained. The recovered code has no relay output pin or pump-control routine, so no actuator wiring map is reconstructed here.

Power and integration

D2 carries the pulse signal; A4 and A5 provide SDA and SCL on the Uno. Confirm module supply ratings and signal compatibility before any future reuse. A pump requires a rated external supply and switching/protection arrangement. Keep electronics dry and supported away from the water path.

3. Source review and setup

The project can be reviewed without a board. These steps also describe future hardware reuse; they do not require reconstructing the exhibit.

Step	Action
1. Open the sketch	Extract the project. Open firmware/AquaFlush_Original_Recovered/ AquaFlush_Original_Recovered.ino in Arduino IDE.
2. Select the platform	Install Arduino AVR Boards if needed; select Arduino Uno. A port is not needed just to compile.
3. Add the LCD library	Install a LiquidCrystal_I2C implementation with the three-argument constructor and init() API. Wire.h comes with the board platform.
4. Review / compile	Use Verify. A fresh compilation was not performed for this revision, so build success is not claimed.
5. Optional hardware reuse	Confirm ratings and wiring, connect a USB data cable, choose the board port, and Upload. Open Serial Monitor at 9600 baud.

Dependency record

Original IDE, board-package, and LCD-library versions were not recorded. The johnrickman/LiquidCrystal_I2C reference source exposes the API used here; its inspected metadata reports version 1.1.4 and author Frank de Brabander. This is an API reference, not an identified historical dependency or a newly verified build.

Output described by the source

When	LCD	Serial
Initialization	Flow Monitor	Serial starts at 9600 baud
Approximately every second	Flow: <integer> L/h	<integer> L/hour

Angle-bracket text describes the format; it is not a captured result. No pump-status line is printed by this sketch.

Troubleshooting for future reuse

Symptom	Check
Missing LCD header / API error	Install the intended library and compare its constructor/init() API.
Blank LCD	Check rated supply, contrast, SDA/SCL, and the assumed address 0x27.
Zero or unstable flow	Check D2, common ground, sensor requirements, and source conditions. Calibration and counter handling also limit this implementation.
Pump does not respond	No pump-control routine is present in the recovered sketch.

4. Prototype photographs

These photographs are retained from the uploaded archive. Its catalog attributes them to the May 2025 presentation; the slide deck was not included in the archive.

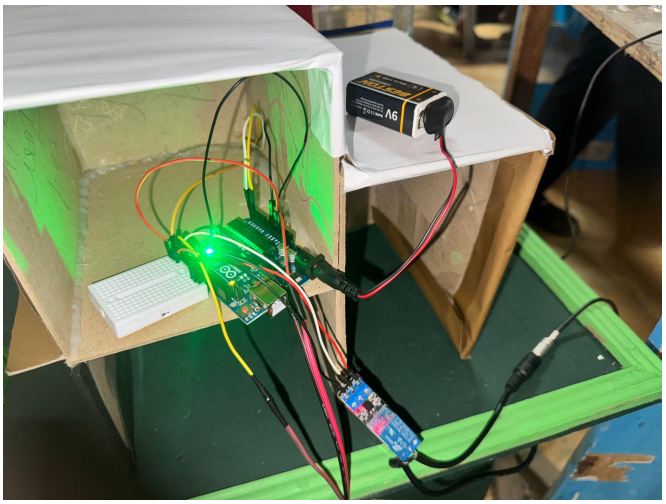


Figure 5. Electronics and relay module inside the scale model.



Figure 6. LCD placement at the front of the model.

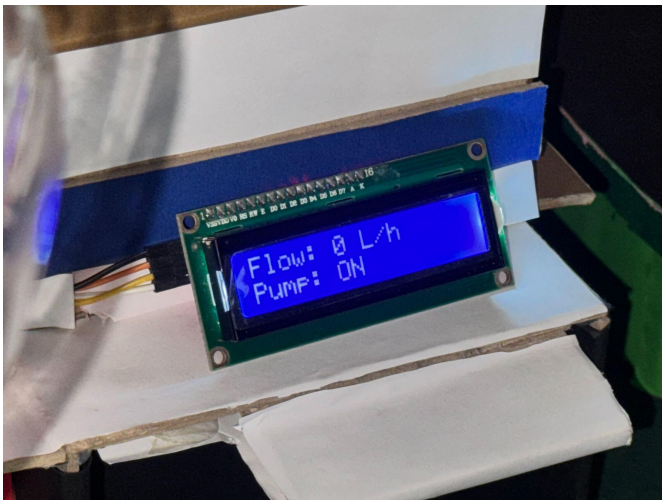


Figure 7. LCD close-up with flow and pump-status label.



Figure 8. Water container and circulation hardware beneath the exhibit.

Photograph/source note: The LCD close-up shows a pump-status message, but the recovered sketch has no corresponding display text or actuator routine. The available files do not resolve that discrepancy; the image records the exhibit rather than establishing automatic pump behavior.

Construction record

The supplied images document a miniature roadway, drainage piping, a mounted character display, internal electronics, and a water container/circulation arrangement. They do not establish exact module ratings or a complete circuit map. The repository photo gallery retains all five original files and their cataloged slide references.

5. Scope, attribution and development

Area	Available record
Implemented monitoring	Pulse acquisition, estimated L/hour, LCD text, and Serial output are present in recovered source.
Exhibit hardware	Scale model, relay, and pump are documented photographically.
Proposed extensions	Automatic actuation, clog classification, and network telemetry are absent from recovered source.
Performance record	Sensor calibration and quantitative accuracy are not established by the uploaded materials.

Source provenance

The supplied notes state that the lost original sketch was reconstructed from a presentation screenshot and identified by a project member as the only firmware used. The slide deck was not supplied for independent review. The recovered sketch is preserved unchanged in its matching Arduino folder under firmware/. This portfolio edition is maintained by Phoebien D. Cañete, with original group attribution retained.

Team contributions

Team member	Reported contribution
Phoebien D. Cañete	Arduino programming, wiring, and prototype construction
Dja-ver Q. Hassan	Arduino programming, wiring, and prototype construction
Tristan E. Listanco	Arduino programming, wiring, and prototype construction
Mary Grace V. Soronio	Arduino programming, wiring, and prototype construction
Shanice Reih H. Tanque	Arduino programming, wiring, and prototype construction

Future development

Potential iterations include recovering component identities, improving the hydraulic layout and measurement code, adding complementary sensing, developing supported obstruction detection, and then introducing rated actuator control and connectivity. These are proposed developments, not completed features.

References

1. Supplied AquaFlush archive, recovered sketch, photographs, and earlier report. Original slide deck not supplied.

2. Arduino. UNO R3 documentation and full pinout.
https://docs.arduino.cc/hardware/uno-rev3/

3. Arduino. Sketch specification.
https://docs.arduino.cc/arduino-cli/sketch-specification/

4. Arduino. Installing a library in Arduino IDE 2.
https://docs.arduino.cc/software/ide-v2/tutorials/ide-v2-installing-a-library/

5. johnrickman. LiquidCrystal_I2C source and metadata.
https://github.com/johnrickman/LiquidCrystal_I2C

Technical sources reviewed 22 October 2026. Full technical source notes are included in docs/REFERENCES.md.